Mod08 Clustering

Unsupervised Learning

So far, we have been doing supervised learning we had features and examples: each of those samples had labels on it (some sort of class, a number, a category; using those we were able to use those labels to predict labels for new examples/points the we haven't seen yet.

Now we move into unsupervised learning.

We now have bunch of unlabeled data.

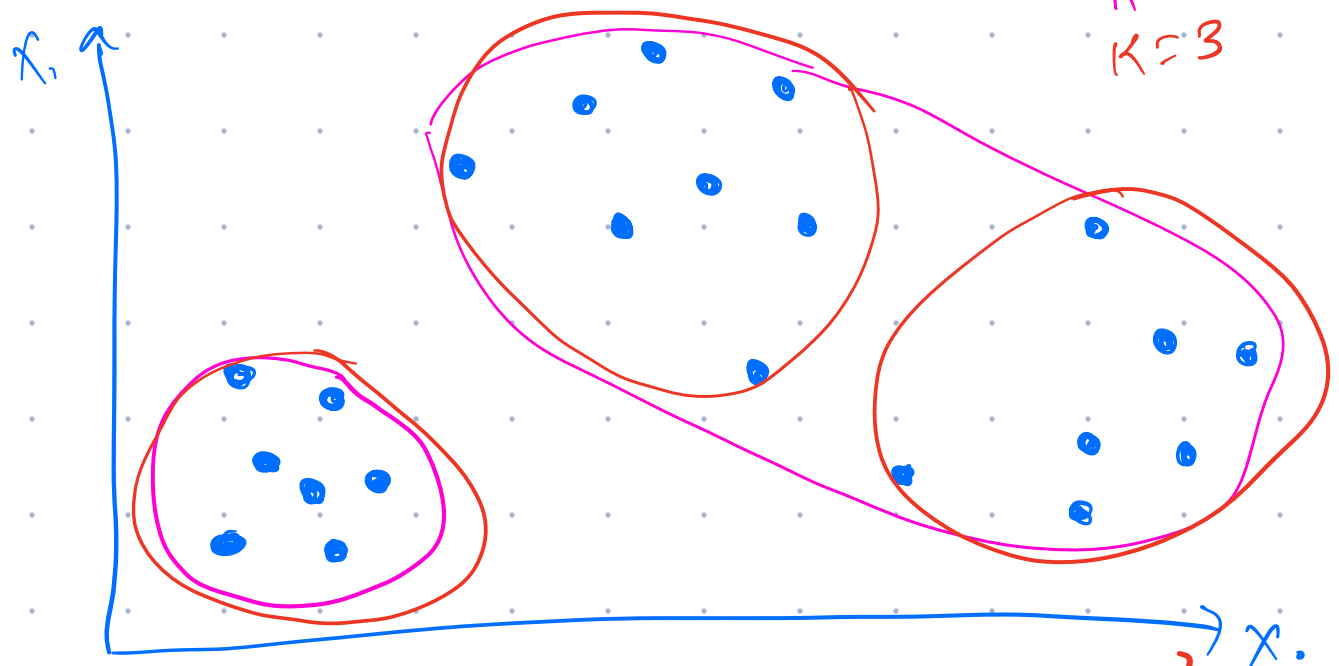
Now what can we do here?

Can we learn anything from this data?

K-means clustering

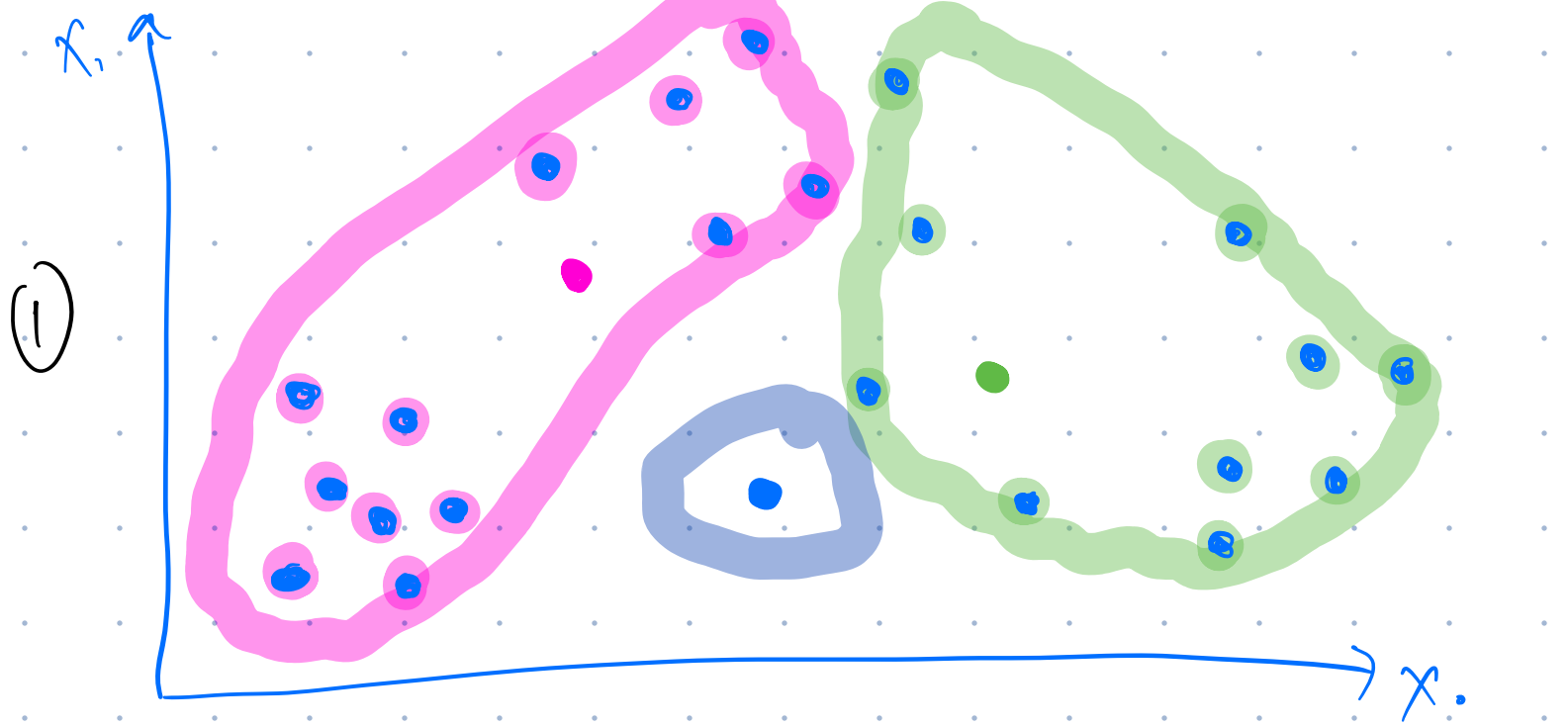
- compute K clusters from data
  K is predefined by you

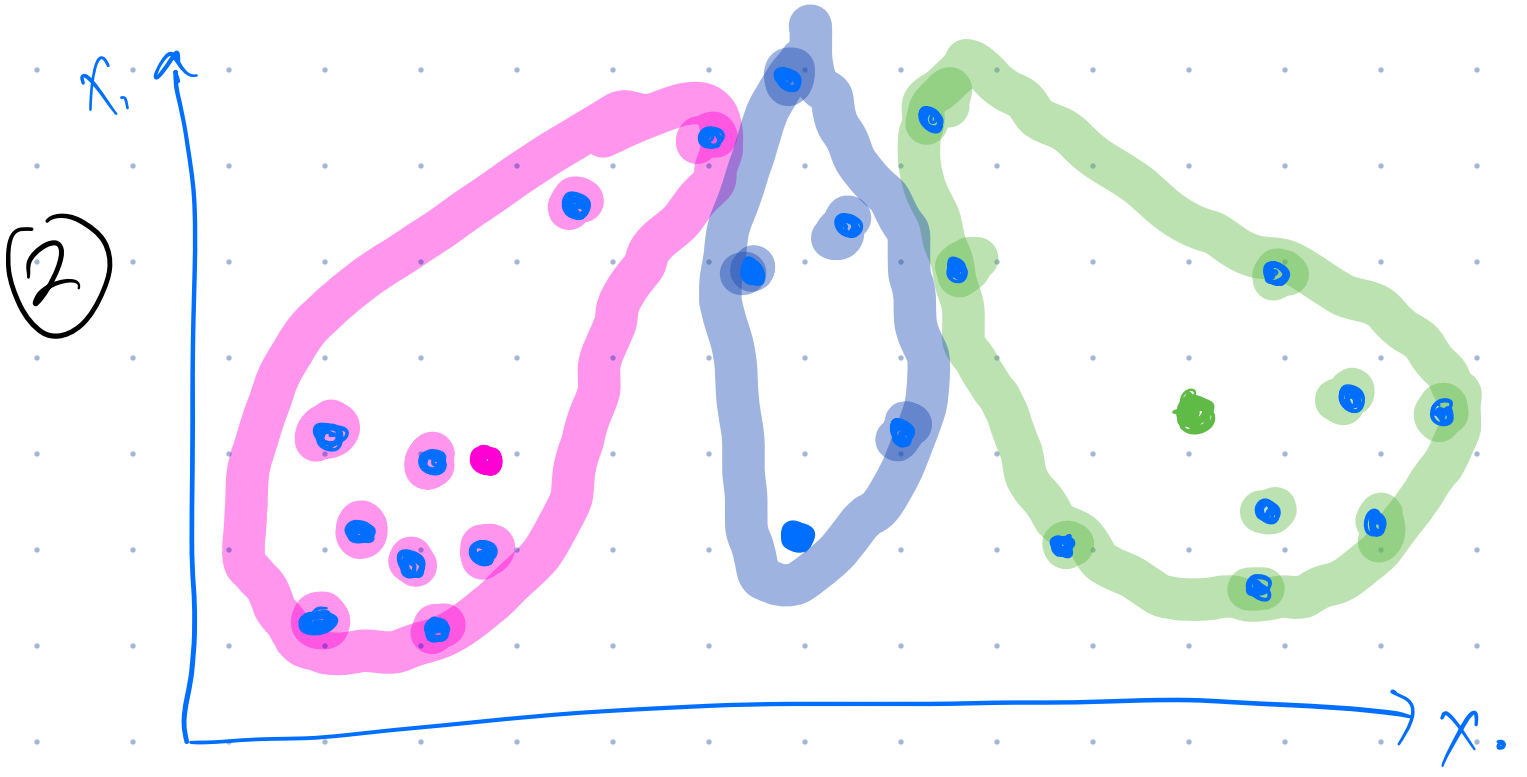
K = 2
K = 3

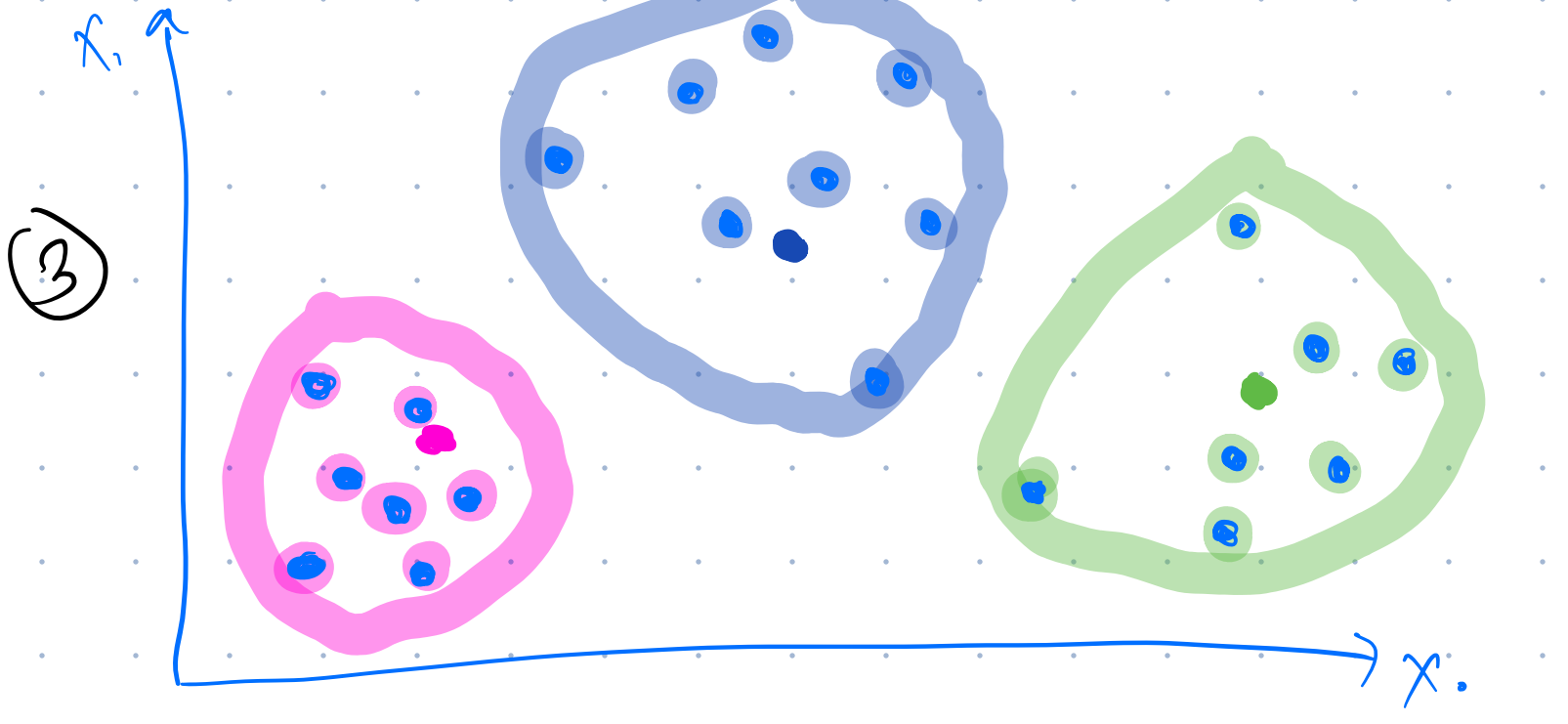


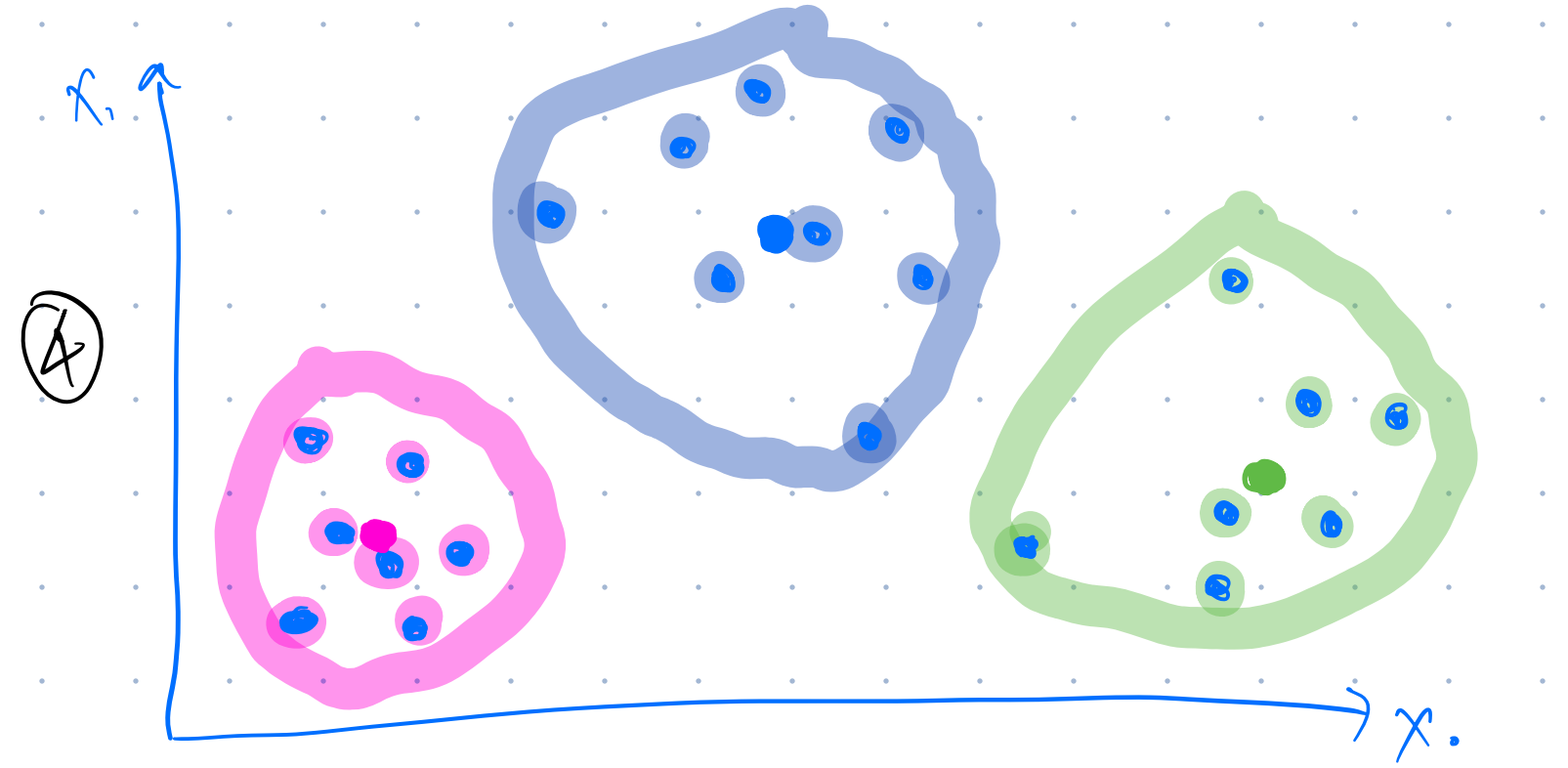
K-means Clustering

1. Choose 3 random points to be the centroid
2. Calculate distance betⁿ points and centroid, assigning points to closet centroid.
   Expectation step.
3. Compute new centroids.
   Maximization


①
x_1
x_2


②
x_1
x_2





None of these 3 cluster (centroids) are changing with further calculation

we go to the user and say, hey these are our three clusters.

Hierarchical Clustering

1. Start by assigning each item to a cluster, so that if you have N items, you now have N clusters, each containing just one item.

2. Find the closest (most similar) pair of clusters and merge them into a single cluster, so that you now have one fewer cluster.

3. Continue the process until all items are clustered into a single cluster of size N.

Linkage metrics?

1. Single linkage:

Consider distance betⁿ one cluster to another cluster to be equal to the shortest distance from any member of one cluster to the any member of other cluster.

2. Complete linkage:

greatest distance from any member of one cluster to any member of the other cluster.

3. Average linkage:

average distance

Example of City-distance

	BOS	NY	CHI	DEN	SF	SEA
BOS	0	206	963	1949	3095	2979
NY		0	802	1771	2934	2815
CHI			0	966	2142	2013
DEN				0	1235	1307
SF					0	808
SEA						0

1. {BOS} {NY} {CHI} {DEN} {SF} {SEA}
2. {BOS, NY} {CHI} {DEN} {SF} {SEA}
3. {BOS, NY, CHI} {DEN} {SF} {SEA}
4. {BOS, NY, CHI} {DEN} {SF, SEA}
5. {BOS, NY, CHI, DEN} {SF, SEA}

→ Single linkage (shortest BT).

or

6. {BOS, NY, CHI} {DEN, SF, SEA}

→ Complete linkage (largest)